F-2021

Masum - 50

3 (a) $\Delta_o$ value . Repeat Q.

(b,c) ~~Bond~~ Chapter

(d) What is crystal field stabilization energy? Explain the splitting ($CFSE_{oct}$) of d orbitals under octahedral

Ans:

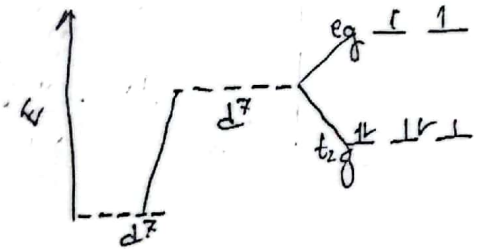
CFSE: an electron is distributed in $e_g$ and $t_{2g}$, we can calculate the energy This energy is called CFSE.

Next part: Already discussed.

(e) Find out the CFSE for a $d^7$ ion, both as a high spin and as a low-spin in an octahedral complex. (2)

soln:

For high spin, the splitting $d^7$ orbitals as shown below —



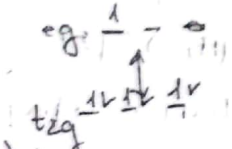

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$CFSE = (-0.4\times5 + 0.6\times2)\Delta_o$

$= (-2.0 + 1.2)\Delta_o$

$= -0.8\Delta_o$

For low spin




$CFSE = (-0.4\times6 + 0.6\times1)\Delta_o$

$= (-2.4 + 0.6)\Delta_o$

$= -1.8\Delta_o$

4 a, b repeat Q.

(c) en forms stable chelate with $Cu^{2+}$ ion, yet $[Cu(en)_3]^{2+}$ ion is highly unstable — Explain.

soln:

Step by step reaction is —

$[Cu(H_2O)_6]^{2+} \xrightarrow[K_1]{+en} [Cu(H_2O)_4en]^{2+} + 2H_2O$

$[Cu\underset{en}{(H_2O)_4}] \xrightarrow[K_2]{+en} [Cu(en)_2(H_2O)_2]^{2+} + 2H_2O$

$[Cu(en)_2(H_2O)_2]^{2+} \xrightarrow[K_3]{+en} [Cu(en)_3]^{2+} + 2H_2O$

(en) = Ethylenediamine

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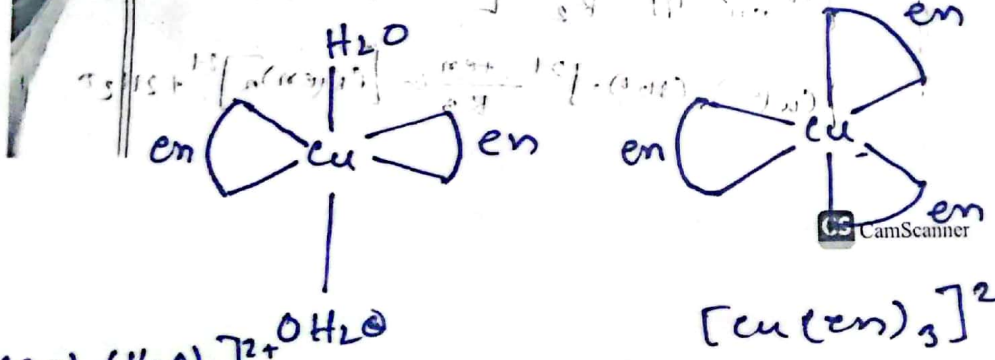
Here, order is $K_1 \approx K_2 >> K_3$

So, $[Cu(en)_3]^{2+}$ is unstable.

According to the Jahn-Teller distortion-

$[Cu(en)_2(H_2O)_2]^{2+}$ occurs distortion in its octahedral shape. 2 mole of $H_2O$ removes and here no distortion occurs in en.

On the other hand, it is not possible ~~without distorting~~ of distortion of octahedral shape without distorting the two chelate ring in the $[Cu(en)_3]^{2+}$ complex. It is not necessarily possible to increase the distance between the atoms of en which are coordinated. As a result $[Cu(en)_3]^{2+}$ complex is highly unstable.




$[Cu(en)_2(H_2O)_2]^{2+}$ $[Cu(en)_3]^{2+}$.

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Here, order is $K_1 \approx K_2 >> K_3$

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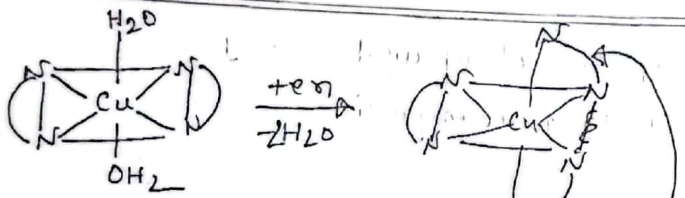
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SUZON-50 Unexpected distortion.

5. (a) Complexes of monovalent silver are diamagnetic, bivalent Ag are paramagnetic and trivalent Ag are again diamagnetic — explain.

Ag (47) : [Kr] $4d^{10}$ $5s^1$

Monovalent complex of Ag :

$Ag^+$ : $4d^{10}$ | 5s | 5p

Ex: $Ag(NH_3)_2$

where No unpaire $e^-$ so diamagnetic.

Divalent complex of Ag

$Ag^{2+}$ : $4d^9$

Here, No. of upaired $e^-$ = 1

so, it is paramagnetic

$Ag^{3+}$

$Ag^{3+}$ : 4d

With the presence of ligand, rearrangement occurs.

$Ag^{3+}$ :

No unpaired $e^-$ here,

so it is diamagnetic.

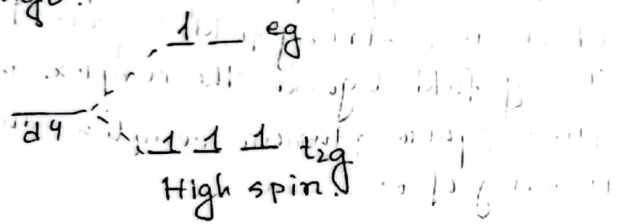
(b) The high spin d4 complex $[Cr(H_2O)_6]^{2+}$ is labile, but the low-spin $d^4$ complex ion $[Cr(CN)_6]^{4-}$ is inert. Clarify this statement.

soln Ligand field strength consider and Cr

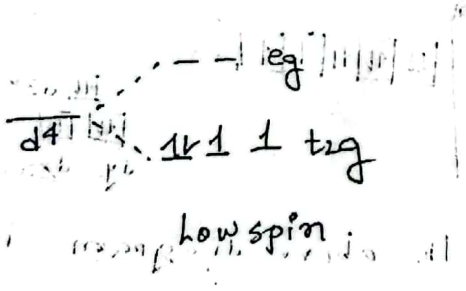
The key factor is the strength of the ligand field.

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Weak field ligands like $H_2O$ lead to high spin complexes with more unpaired e- and In this case, electrons occupy both lower energy $t_{2g}$ and $e_g$ (higher energy) making them more accessible for ligand to exchange.




While strong field ligands like $CN^-$ lead to low spin complexes with fewer unpaired e- and higher inertness.

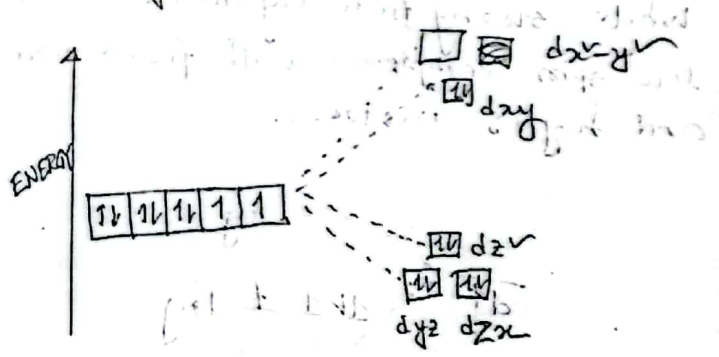



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(d) The $[Ni(CN)_4]^{2-}$ ion, which has a square planar geometry, is diamagnetic, whereas $[NiCl_4]^{2-}$ ion, has a tetrahedral, is paramagnetic. Show CFS diagram.

In $[Ni(CN)_4]^{2-}$, the ligand present is $CN^-$ which is a strong field ligand. Due to the strong field ligand, the complex is a low spin – square planar complex due to the pairing of e-.

The CFS diagram is —




In the above diagram, it is observed

$[Ni(CN)_4]^{2-} \longrightarrow Ni^{2+}$

$Ni^{2+} \longrightarrow [Ar]\ 3d^8$

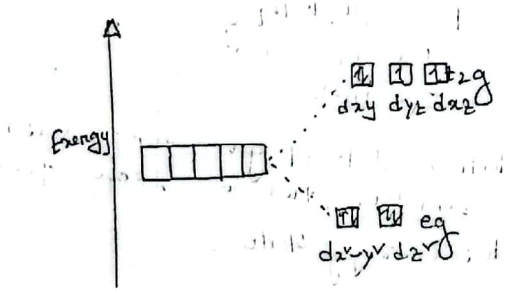
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that all the e⁻ are paired therefore the complex is diagmagnetic.

In $[NiCl_4]^{2-}$, the present ligand is $Cl^-$ which is a weak field ligand, so the complex is a high spin - tetrahedral complex

The CFS diagram is,

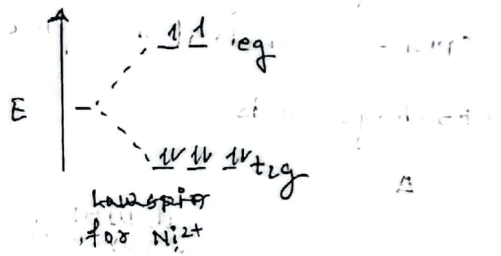



Here, unpaired $e^-$ = 2

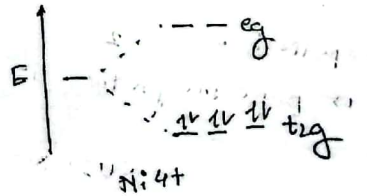
So, it is paramagnetic in nature.

8 (a) $K_2[NiF_6]$ is diamagnetic while $K_3[CoF_6]$ is paramagnetic though both have same $d^6$ configuration.

Soln:




The oxidation state of given compound is +4. So another two $e^-$ leave from the higher energy state.




Here, we can see that no unpaired $e^-$

So ~~$K_3[CoF_6]$~~ i $K_2[NiF_6]$ is diamagnetic

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On the other hand,

$Co^{3+}$ $\underline{1}\ \underline{1}$ $e_g$

$d_{z^2}$

$\underline{1\downarrow}\ \underline{1\downarrow}\ \underline{1\downarrow}$ $t_{2g}$

for, $d^6$ $\underline{1}$ — $e_g$

$\underline{1\downarrow}\ \underline{1\downarrow}\ \underline{1}$ $t_{2g}$

Here unpaired e⁻ is present, so paramagnetic

(b) Repeat

(c) Explain why CFSE for a low spin $d^5$ configuration contains 2P term.

Soln:

When a $d^5$ complex is a low spin state, all five electrons occupy the lower energy $t_{2g}$ orbitals, forcing two electrons to pair up.

Pairing energy (P): The energy required to pair two e⁻ in the same orbital.

When calculating CFSE for $d^5$ complex, we need to consider the energy gained by placing e⁻ in the lower $t_{2g}$ orbitals, which is then reduced by the energy cost of pairing the two electrons, resulting in a '2P' term in the equation.

(b) The absorption of $[Ti(H_2O)_6]^{3+}$ exhibits a bands with $\lambda_{max}$ = 510 nm. What color of light absorbed and what color will aq solutions of $[Ti(H_2O)_6]^{3+}$ appear? Why?

→ 500 – 560 nm (Green)

Soln:

$Ti^{3+} \rightarrow 3d^1$

— — $e_g$ ↑ —

↑ — — $t_{2g}$ — — —

Ground state of $Ti^{3+}$ ($3d^1$) Excited state

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$Ti^{3+}$ absorb green[ish-yellow] photon ($\lambda_{max} = 510\,nm$) to excite electron to higher energy level.

The $[Ti(H_2O)_6]^{3+}$ ion absorbs light in the greenish yellow region, meaning that when a substance absorbs greenish-yellow light, it will appear purple. So the aq. soln is purple in color

SUZON-50

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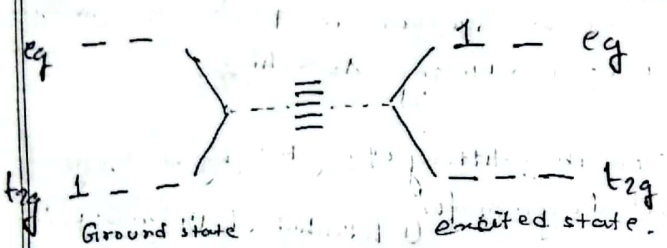
5 (a) Define crystal field splitting. How would you measure it experimentally? (2)

The separation of fived-orbitals of the metal ion into two sets having different energies is called crystal field splitting.

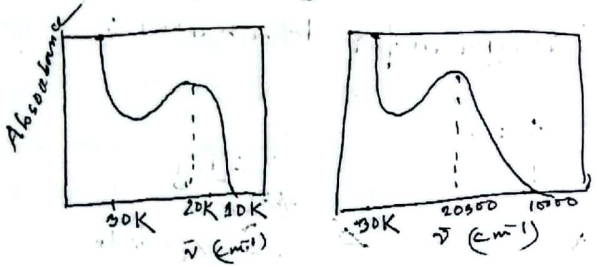
The value of crystal field splitting $\Delta_o$ betn $e_g$ and $t_{2g}$ can be measured easily by recording the UV-visible spectrum of the complex. Consider a complex like $[Ti(H_2O)_6]^{3+}$ The $Ti^{3+}$ ion has one d electron. In the complx this will occupy the orbital with the lowest energy, that is one of the $t_{2g}$ orbitals. The complex absorbs lights of the correct wavelength to promote the electron from the $t_{2g}$ level to the $e_g$ one

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Spectrum for $[Ti[H_2O]_6]^{3+}$ is



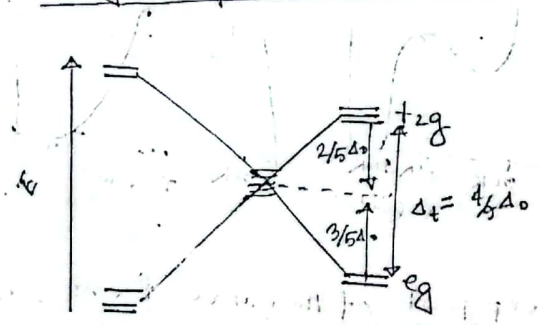

The steep part of the curve from 27000 to 30000 is due to the charge transfer. The d-d transition is single broad peak with a max. 20,300 $cm^{-1}$. Since, $1 KJ mol^{-1} = 83.7 cm^{-1}$ so, $\Delta_o$ is $\frac{20300}{83.7} = 243$ KJ/mol. This value is much the ~~so~~ same as the energy of many normal single bonds

The above method is the most convenia way of measuring $\Delta_o$ values.

(b) Show the splitting of any $M^{2+}$ by energy level diagram for (I) Tetrahedral (II) squareplanar ligand env.

Ans:

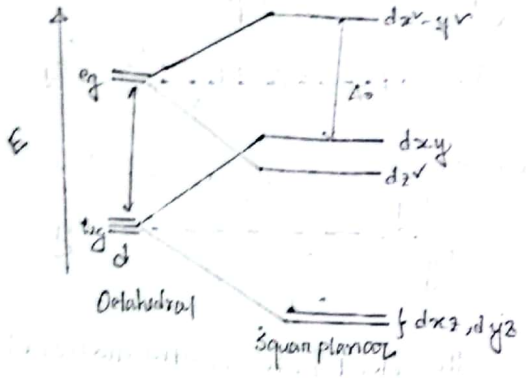
Diagram for Tetrahedral:




Octahedral Tetrahedral

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Diagram for Squar planar:




(a) Define CFSE. Calculate CFSE for $[V(H_2O)_6]^{3+}$ in KJmol$^{-1}$. Given $\Delta_o = 18900$ KJ mol$^{-1}$.

Soln:

CFSE: The d-orbitals of the ion in the field of spherical symmetry are split in an octahedral field. The energy achieved by preferential filling of lower lying $t_{2g}$ orbitals is called the crystal FSE.

(Since it stabilizes d-orbitals by lowering their energy, which results from splitting.

into $t_{2g}$ and $e_g$ CFSE can be calculated by the following equation)

$$CFSE = \left(-\tfrac{2}{5} n_{t_{2g}} + \tfrac{3}{5} n_{e_g}\right) \Delta_o$$

where,

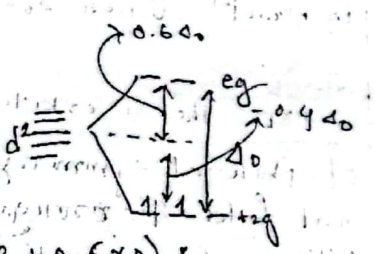
$n_{t_{2g}}$ = no. of e$^-$ in $t_{2g}$

$e_g$ = no. " " " $e_g$

Let the electronic configuration of V is-

$V^{3+}(23)$ : $\sigma$ [Ar] $3d^2\ 4s^0$

Splitting of d2




Now, CFSE = $(-0.4 \times 2 + 0.6 \times 0)\Delta_o$

$= -0.8\Delta_o = 0.8 \times 18900$

$= -15120$ kJ mol$^{-1}$

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(d) What is pairing energy? calculate CFSE for high spin $[Fe(H_2O)_6]^{2+}$ complex in KJ/mol. → confuse.

$[\Delta_o = 10,400\ cm^{-1}$, $P = 17,600\ cm^{-1}$, $1\ KJ\ mol^{-1} = 83.7\ cm^{-1}]$

Soln:

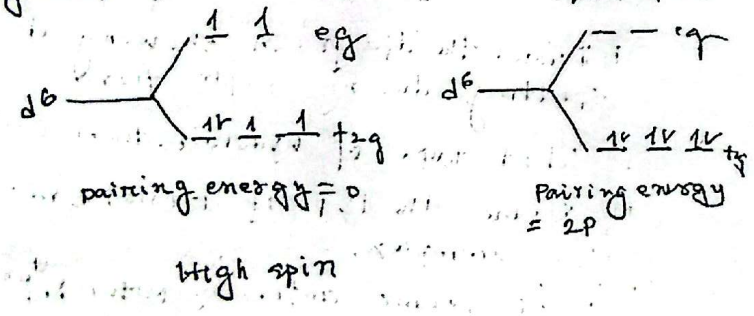
Pairing Energy: The amount of energy required to pair two single electrons in an orbital is known as pairing energy. It is denoted by 'P'.

↑↓ P → ↑ ↑

For octahedral: $\Delta_o < P$ or $\Delta_o > P$ may for H.S or LS complx

For tetrahedral: $\Delta_t < P$ → Always form High spin complex.

For high spin,




For: Bold 45°

Here, $\Delta_o < P$

That indicates that this is high spin comp

Paining doest occurs here,

$\therefore CFSE = \{0.4 \times 4 + 0.6 \times 2\}\ \Delta_o$

$= -0.4\ \Delta_o$

$= -0.4 \times \frac{10400}{83.7}$ [1 KJ mol⁻¹ = 83.7 cm⁻¹]

$= -49.70\ KJ\ mol^{-1}$

If paining occurs we should add $[0.4\ \Delta_o + P]$

a, b → Repeat

6.

(c) A metal ion $M^{n+}$ having $d^4$ combined with three bidentate ligands to form a complex compound. $[\Delta_o > P]$

(i) Draw the diagram showing d- orbital splitting during complx form

(ii) What type of hybridization will $M^{n+}$ have?

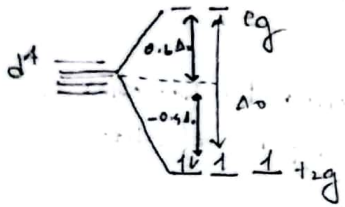
(iii) Name the type of isomerism exhibited by this complex.

(iv) electronic confi. of metal $M^{\oplus n}$.

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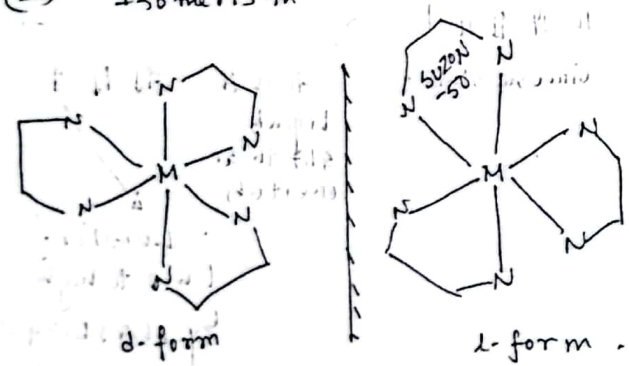
(i) Here, $\Delta_o > P$

So, Low spin complx.




(II) $\Delta_o > P$ → Low spin.
inner orbital complex
Hybridization $d^2sp^3$

(III) Isomerism

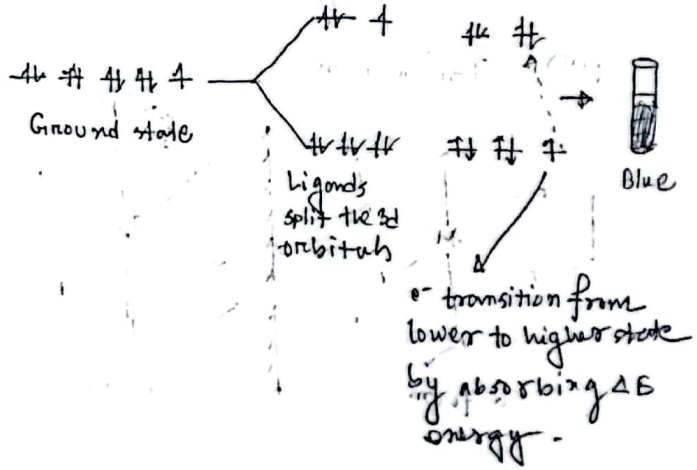



(IV) $Mn^{+}$ (24e⁻) ⇒ $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

(V) How do you explain the blue color of $Cu^{2+}$(aq) solⁿ, whereas $Zn^{2+}$(aq) is colorless.

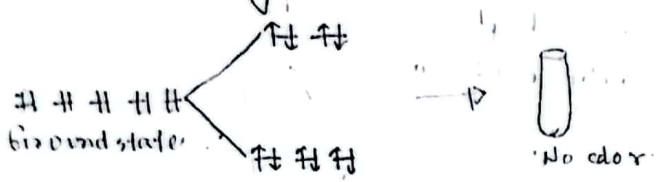
Solⁿ:

$Cu^{2+}$ ion with $H_2O$ ligands — Blue color.




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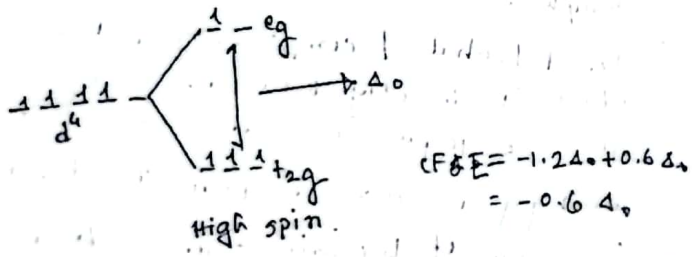
$Zn^{2+}$ with ligands →




(b) $Cr^{2+}$ is very easily oxidized to $Cr^{3+}$ — explain in terms of CFT.

Soln: In weak ligand condition.

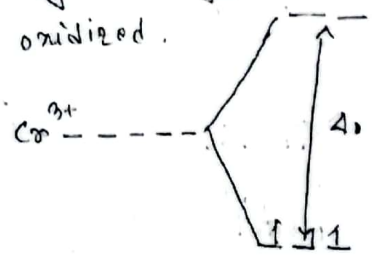
$Cr^{2+} \rightarrow 3d^4$




$CFSE = -1.2\Delta_o + 0.6\Delta_o$
$= -0.6\Delta_o$

Here, CFSE, $\Delta_o$ is low than the with strong $Cr^{3+}$.

So, it is easier to remove an electron from higher energy state. So $Cr^{2+}$ is easily oxidized.




$CFSE = (0.4 \times 3)\Delta_o$
$= -1.2\Delta_o$

(c) Tetrahedral complx are usually of the high spin complex — explain.

Soln: Tetrahedral complexes are generally high spin complex:

In tetrahedral complex, there has no d-orbitals which is directly towards the ligands lowering the e⁻-e⁻ repulsion and therefore splitting of energy is less compare to the octahedral less no. of ligands produce less crystal field splitting.

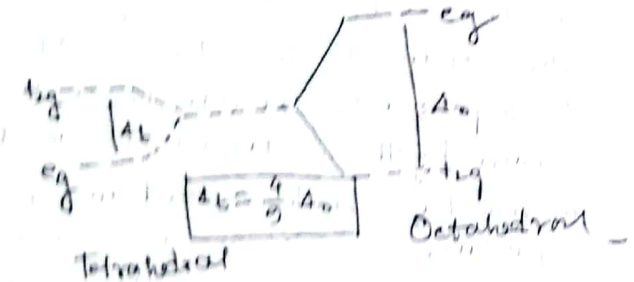
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Since the magnitude of CFSE is small and always less than the pairing energy.

$\Delta_o < P$

So, the pairing of e⁻ will never be energetically favourable. Thus they are high spin complexes.

Also colored, (d–d transition)
Because less energy required.




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[F-2019]

Describe the CFT for transition metal complexes.

Soln:

It is a model describe the breaking of degeneracy of electron orbital states, due to a static electronic field produced by a surrounding charge distribution.

The theory assumes that the attraction betn central metal and the ligand is purely electrostatic. The transition metal of the complex is regarded as a positive charge which is surrounded by the negative or neutral ligands. The electrons on the central metal or under repulsive forces from those on the ligands. Thus the electrons occupy the d- orbitals furthest away from the direction of approach of ligands. The CFT has the following assumptions.

(1) Ligands are treated as point charge.

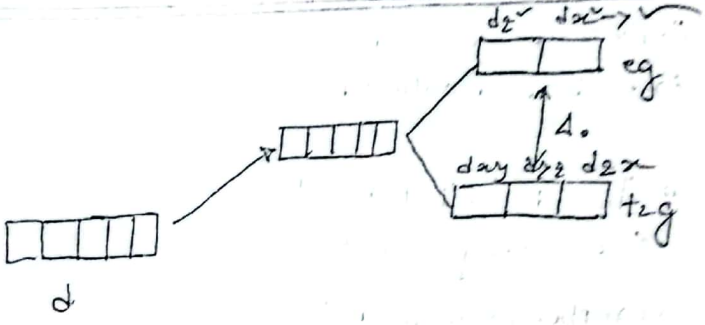
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2. There is no interaction betn metal orbital and ligand orbital.

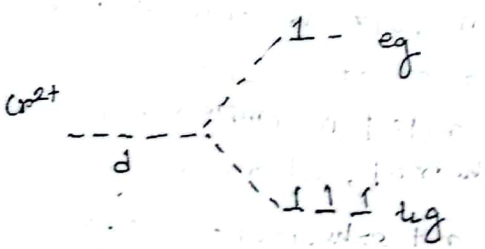
3. The d-orbital on the metal all have the same energy in the free atom. However, a complex is formed the ligands destroy the degeneracy of these orbitals.

For example – If a spherical symmetrical field of negative charges surround the metal ion, the d orbital remain degenerate. However, the energy of the orbitals raised because of repulsion betn the field and e⁻ on the metal. In most cases 6 or 4 ligand surround the metal in complex, giving octa- and tetrahedral, where charge of the ligand is not spherically symmetrical. Thus the d-orbitals are not all affected equally by the ligand e field.




(d) Define high & low spin. Which one is high spin? Justify your answer (2+4)

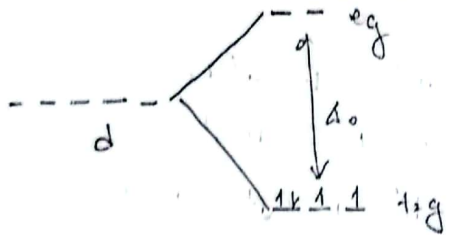
(i) $[Cr(H_2O)_6]^{2+}$ & $[Cr(CN)_6]^{4-}$




so high spin.
Where $H_2O$ weak ligand.

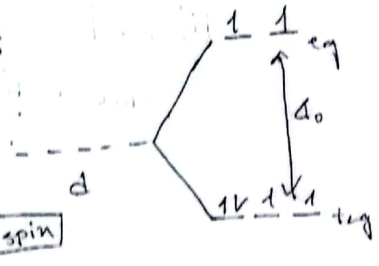
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$Co^{2+}$ with $CN^-$ (Strong ligand)



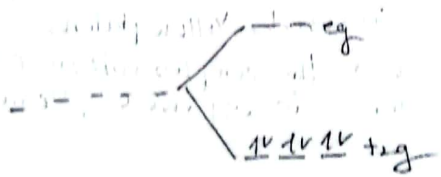

(II) $[Fe(H_2O)_6]^{2+}$

$Fe^{2+} \longrightarrow 3d^6$




High spin

$[Fe(CN)_6]^{4-}$




Low spin

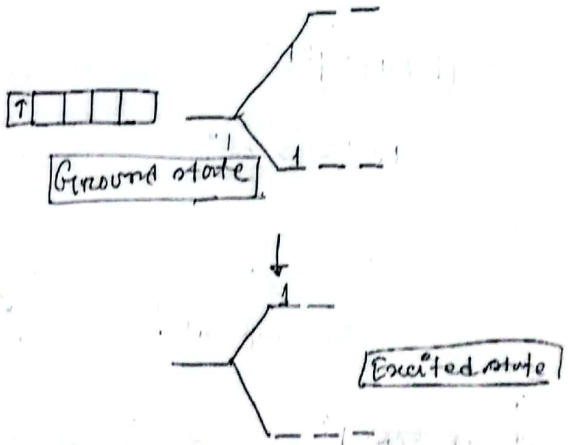
(a) $TiCl_4$ is colorless whereas $[Ti(H_2O)_6]^{3+}$ is violet.

Sol[n]:

$Ti^{4+} \longrightarrow 3d^0 4s^0$

No e⁻ in d

$Ti^{3+} \longrightarrow 3d^1$ No. of e⁻




It absorbs Yellow photons and the complementary color is violet which is appear at our eyes.

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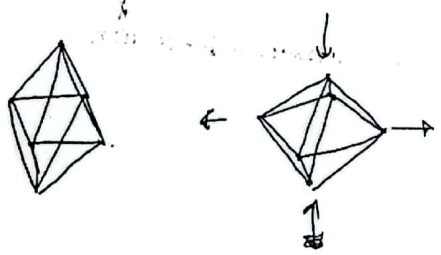
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(c) "Tetragonal distortion" and "rhombic distortion" in octahedral molecules with diagram.

Soln:

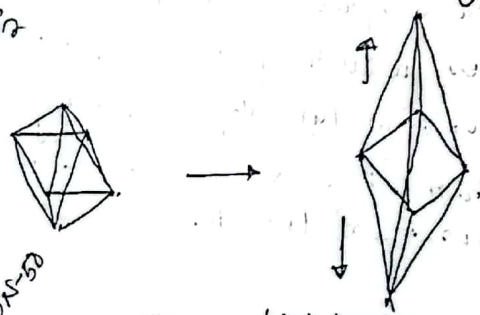
A "rhombic distortion" in a octahedral complex refers to a distortion where the complex deviates from perfect octahedral symmetry by stretching or compression along two opposing diagonal axes, resulting in a shape resembling a rhombus and typically caused by Jahn-Teller effect when there is an uneven distribution of electrons in the d orbitals.



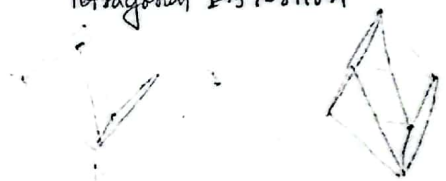
Rhombic distortion

A tetragonal distortion as a special case:

A special case of rhombic distortion is a tetragonal distortion, where the elongation or compression occurs only along one axis, leading to a stretched or compressed octahedron along the z-axis.



Tetragonal Distortion



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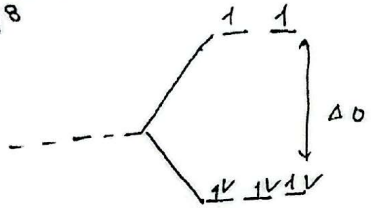
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7. Calculate the CFSE for a $d^8$ ion such as $Ni^{2+}$ in octahedral and tetrahedral complexes. Use the units of $\Delta_o$ in both cases. Which is the most stable?

Sol(n):

Octahedral:

$Ni^{2+} \rightarrow 3d^8$

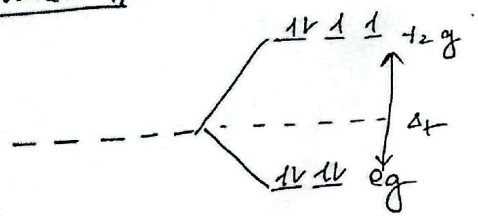



$CFSE = (-0.4 \times 6 + 0.6 \times 2)\Delta_o$

$= (-2.4 + 1.2)\Delta_o$

$= -1.2\Delta_o$

Tetrahedral:




$CFSE = (-0.6 \times 4 + 0.4 \times 4)\Delta_t$

$= (-2.4 + 1.6)\Delta_t = -0.8\Delta_t = -0.8 \times \frac{4}{9}\Delta_o$

$= -0.355\Delta_o$

∴ CFSE of octahedral is greater than the tetrahedral. So octahedral is more stable. where more interactions between the metal ion and ligands.

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